BRIDGE: Balancing Representations by Identifying and Generating Underrepresented Data

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Abstract

Self supervised pretraining inherits the sampling bias of its source data. Most remedies for long tailed self supervised learning are model centric. They change the loss, schedule, or optimization dynamics while leaving the training distribution fixed. We study a data centric alternative. BRIDGE alternates between self supervised pretraining, non parametric OOD scoring in representation space, and targeted image generation with a frozen diffusion model. The final method replaces our earlier extreme tail selector with a histogram based mode window that targets a dense sparse band and avoids the visually noisy far tail. Across ImageNet-100-LT, CIFAR-10-LT, CIFAR-100-LT, PASS, and DiffusionDB source pretraining, BRIDGE achieves the best average linear probe transfer among the methods we evaluate. On ImageNet-100-LT source pretraining, BRIDGE improves average linear transfer over imbalanced SimCLR by 3.05 points. Across the five source regimes, the gain over source matched SimCLR ranges from 2.41 to 4.89 points. Ablations show that Stable Diffusion 3 is the strongest generator, SimCLR is the strongest SSL objective for the repair loop, and the new mode window selector improves over the older top tail rule, although uniform sampling remains competitive in the isolated selection ablation. Full linear and kNN tables, additional supporting figures, runtime statistics, and license notes are provided in the appendix.

1 Introduction

Self supervised learning has become a standard tool for learning transferable visual representations from unlabeled images. Strong methods now span contrastive learning, masked prediction, and self distillation [Chen et al., 2020, He et al., 2020, 2022, Caron et al., 2021, Oquab et al., 2024]. Yet these methods still inherit the sampling bias of the source corpus. When the source data is long tailed or otherwise skewed, the learned embedding allocates most of its capacity to dense majority regions and leaves sparse regions weakly modeled. This hurts transfer, especially on fine grained tasks and tail heavy recognition settings.

Most prior attempts to address long tailed SSL are model centric. Temperature Schedules reshapes the optimization trajectory of contrastive learning over time [Kukleva et al., 2023]. SDCLR introduces a self competitor to emphasize harder and rarer samples [Jiang et al., 2021]. Other work adds external OOD data or frequency aware training heuristics to the SSL loop [Bai et al., 2023, Lin et al., 2023]. These are effective ideas, but they still treat the source distribution as mostly fixed.

We study the complementary direction. Can we repair the source dataset itself while training proceeds? Our answer is BRIDGE, a cyclic data repair loop. After each pretraining stage, we embed the current training set, score each image by mean kNN distance, select a sparse but semantically recoverable region of the representation space, and generate new training images for that region with a frozen diffusion model. The next cycle trains on the expanded dataset. The encoder therefore exposes where the data manifold is thin, and generation densifies those weak regions before the next round of learning.

The submission differs from our earlier draft in two important ways. First, the main method no longer selects the most extreme OOD samples. Instead it builds a histogram over OOD distances and draws a fixed budget around the mode of that histogram. This skips the super OOD tail, which often contains noisy or weakly useful images. The old tail only rule is retained as an ablation. Second, the empirical scope now extends beyond ImageNet derived long tails. In addition to ImageNet-100-LT, CIFAR-10-LT, and CIFAR-100-LT, we evaluate web scale unlabeled source data from PASS and DiffusionDB.

This paper makes three contributions. First, it introduces a data centric SSL loop that repairs sparse representation regions with frozen diffusion augmentation. Second, it replaces pure tail selection with a histogram based mode window that is better aligned with semantically recoverable sparse regions and improves over the older top tail rule. Third, it provides a broad empirical study across five source regimes, seven downstream tasks, three seeds, strong baselines, and targeted ablations.

2 Related Work

Self supervised visual representation learning. Modern vision SSL is dominated by contrastive methods such as SimCLR and MoCo [Chen et al., 2020, He et al., 2020], self distillation methods such as DINO [Caron et al., 2021], and masked prediction methods such as MAE [He et al., 2022]. These methods have made unlabeled pretraining routine, but they usually take the source dataset as given.

Long tailed and imbalanced SSL. Long tailed data changes both optimization and representation geometry. TS addresses this with a schedule over the contrastive temperature [Kukleva et al., 2023]. SDCLR prunes a self competitor to emphasize hard and tail examples [Jiang et al., 2021]. FASSL introduces frequency aware SSL training [Lin et al., 2023]. COLT injects external OOD data into long tail SSL [Bai et al., 2023]. Our setting is different. BRIDGE does not modify the core SSL loss and does not require a separate external OOD collection. It repairs sparse regions using images generated from the training data already in hand.

OOD scoring and data curation. Data centric AI emphasizes that dataset composition is often as important as model design [Whang et al., 2023]. For our setting, the scoring rule must work without labels, density estimation assumptions, or class frequency estimates. Mean deep nearest neighbor distance satisfies these constraints and has been effective for OOD detection in high dimensional representation spaces [Sun et al., 2022].

Diffusion based augmentation. Diffusion models enable high quality image synthesis with strong semantic control [Ho et al., 2020, Rombach et al., 2022]. Prior long tail work has often used generated images in supervised settings. Our use is different. A frozen image to image diffusion model serves as a repair operator inside an SSL loop. Generation is targeted at sparse representation regions instead of globally sampled prompts.

3 Method

3.1 Setting and repair loop

Let $D^{(0)} = \{x_i\}_{i=1}^{N_0}$ denote an unlabeled source dataset and let f_θ be an SSL encoder trained with objective $\mathcal{L}_{\text{ssl}}$. BRIDGE alternates between pretraining and dataset repair over C cycles. At cycle c, the current encoder is $f_{\theta^{(c)}}$ and the current source set is $D^{(c)}$.

After training on $D^{(c)}$, we embed every image and compute a non parametric OOD score from mean kNN distance,

$$z_i^{(c)} = f_{\theta^{(c)}}(x_i), \qquad d_i^{(c)} = \frac{1}{k}\sum_{j=1}^{k} \left\| z_i^{(c)} - \pi_j\left(z_i^{(c)}\right) \right\|_2^2, \tag{1}$$

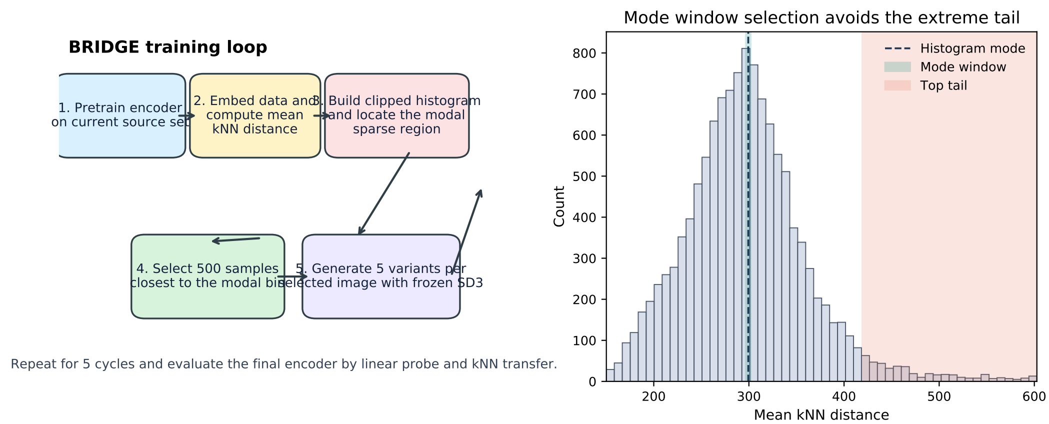

Figure 1: Left: the BRIDGE repair loop. Right: the new mode window selector on a representative OOD histogram. The new selector targets the densest sparse band instead of the extreme right tail.

where $\pi_j\left(z_i^{(c)}\right)$ is the jth nearest neighbor of $z_i^{(c)}$ inside the current training set. Large values indicate that an image lies in a sparse part of the learned representation space.

3.2 Mode window OOD selection

The old version of our method selected the largest OOD scores directly. That rule is simple but brittle. The farthest samples are often visually noisy, outlying even relative to the sparse regions that are still semantically useful for repair.

Our current selector therefore works in two steps. First, it clips the empirical OOD distribution to its 1% to 99% quantile range and builds a histogram over the clipped values. Let $b^\star$ denote the modal bin and let $m^{(c)}$ denote the center of that bin. Second, given an augmentation budget B, we select the B samples whose OOD scores are closest to $m^{(c)}$,

$$S^{(c)} = \arg\operatorname{top-}B_i \; -\left| d_i^{(c)} - m^{(c)} \right|. \tag{2}$$

This is a symmetric selection around the densest sparse region rather than a direct grab of the farthest tail. Figure 1 illustrates the mechanism, and Appendix Figure 4 shows that the OOD shoulder can move substantially across training.

3.3 Diffusion based repair

Given the selected set $S^{(c)}$, we generate G synthetic variants per selected image with a frozen image to image diffusion model g_ϕ. In the main method we use Stable Diffusion 3. The generation ablation replaces it with FLUX or with a no generation re population baseline. The repaired source set is

$$D^{(c+1)} = D^{(c)} \cup \left\{ g_\phi(x, \epsilon_j) \mid x \in S^{(c)},\, j = 1, \ldots, G \right\}. \tag{3}$$

Because the generator is frozen, the learned component remains the SSL encoder. The repair loop is therefore compatible with any SSL objective that can provide embeddings for OOD scoring.

3.4 Design rationale

The key choice is that we do not try to densify the most isolated samples at all cost. The farthest tail often contains degenerate crops, artifacts, or images whose semantics are too unstable for reliable image to image augmentation. The histogram mode captures a more useful target. It finds images that are underrepresented relative to the main body of the representation space, yet still belong to a coherent local neighborhood. In the completed selection ablation, this change improves average linear transfer by 0.32 points and average kNN transfer by 0.77 points over the older top tail rule on ImageNet-100-LT. Uniform sampling remains competitive in that isolated ablation, so we interpret the mode window as a better replacement for pure tail selection rather than as a universally dominant selector.

Table 1: Linear probe comparison between balanced pretraining, imbalanced pretraining, and BRIDGE on ImageNet-100-LT. The reported average is over all seven downstream linear-probe tasks, including Cars and Aircraft.

Method	C10	C100	Flowers	Pets	IN100-LT	Avg.
Balanced	65.3 ± 1.2	37.0 ± 0.8	22.4 ± 1.6	$\mathbf{31.2 \pm 2.0}$	39.0 ± 1.0	31.1 ± 1.1
Imbalanced	65.4 ± 4.2	38.6 ± 1.4	23.7 ± 1.0	28.4 ± 1.3	39.6 ± 0.3	31.2 ± 1.4
BRIDGE (ours)	$\mathbf{70.7 \pm 1.7}$	$\mathbf{43.3 \pm 0.6}$	$\mathbf{26.6 \pm 2.7}$	$\mathbf{31.2 \pm 3.7}$	$\mathbf{42.0 \pm 1.2}$	$\mathbf{34.3 \pm 1.8}$

4 Experimental Setup

Source datasets. We pretrain on five unlabeled source regimes. Three are long tailed image classification datasets, namely ImageNet-100-LT, CIFAR-10-LT, and CIFAR-100-LT. Two are web scale unlabeled corpora, namely PASS [Asano et al., 2021] and DiffusionDB [Wang et al., 2023]. For PASS and DiffusionDB we use 10k example subsets and override the pretraining split to $[1.0, 0.0, 0.0]$, so all sampled images are used for SSL pretraining.

Downstream evaluation. We evaluate transfer on seven benchmarks: CIFAR-10-LT, CIFAR-100-LT, Stanford Cars [Krause et al., 2013], FGVC Aircraft [Maji et al., 2013], Oxford Flowers [Nilsback and Zisserman, 2008], Oxford-IIIT Pets [Parkhi et al., 2012], and ImageNet-100-LT. The main paper reports linear probe results. Full linear probe and kNN tables are deferred to the appendix.

Training protocol. All reported numbers are mean $\pm$ standard deviation over three seeds. The main backbone is ResNet-50. Unless noted otherwise, BRIDGE uses $k = 100$ for OOD scoring, selects 500 images per cycle, generates 5 images per selected image, and trains for 5 cycles with 100 epochs per cycle. The cycle ablation changes the number of cycles while adjusting the per cycle budget accordingly. The successful DINO ablation uses two local crops. Appendix B gives additional implementation details, runtime summaries, and asset notes.

Compared methods. We compare SimCLR, TS, SDCLR, BRIDGE, and two hybrids that combine BRIDGE with TS or SDCLR. The combinations are natural because BRIDGE changes the data while TS and SDCLR change the optimization dynamics.

5 Results

5.1 Balanced and imbalanced baselines

Table 1 compares balanced pretraining, imbalanced pretraining, and BRIDGE on ImageNet-100-LT source data. BRIDGE improves the seven task average linear probe score by 3.05 points over the imbalanced baseline and by 3.14 points over the balanced baseline. The gains are broad rather than isolated. Relative to imbalanced SimCLR, BRIDGE improves CIFAR-10-LT by 5.37 points, CIFAR-100-LT by 4.71 points, Flowers by 2.83 points, and ImageNet-100-LT transfer by 2.35 points.

5.2 Comparison across source datasets

Table 2 reports the average linear transfer over all seven downstream tasks for each source regime. BRIDGE achieves the best average linear transfer in all five source regimes that we evaluate. Relative to source matched SimCLR, the average gain is +2.41 points on ImageNet-100-LT, +3.15 on CIFAR-10-LT, +4.89 on CIFAR-100-LT, +3.15 on PASS, and +2.54 on DiffusionDB. Appendix Figure 2 visualizes the same trend.

The detailed pattern is also informative. Plain BRIDGE is the strongest average transfer method in every regime. BRIDGE+TS is the strongest method on the in domain ImageNet-100-LT target when the source is PASS or DiffusionDB, but those target gains do not translate into the best overall average transfer. SDCLR and BRIDGE+SDCLR are consistently weaker in our setting.

5.3 Ablations

Table 3 summarizes the most important ablations. Three conclusions are stable.

Table 2: Average linear-probe transfer over all seven downstream tasks. The full per-task linear and kNN tables are deferred to the appendix.

Source	SimCLR	TS	SDCLR	BRIDGE	BRIDGE+TS	BRIDGE+SDCLR
ImageNet-100-LT	39.8 ± 1.5	39.6 ± 1.1	15.1 ± 0.8	$\mathbf{42.3 \pm 1.0}$	35.7 ± 1.0	15.2 ± 0.8
CIFAR-10-LT	29.1 ± 0.9	26.2 ± 1.3	13.8 ± 1.1	$\mathbf{32.2 \pm 1.6}$	28.5 ± 0.9	13.3 ± 0.9
CIFAR-100-LT	23.8 ± 1.4	21.9 ± 1.2	13.2 ± 0.7	$\mathbf{28.7 \pm 1.8}$	27.1 ± 2.6	13.2 ± 1.0
PASS-10k	36.8 ± 0.8	37.3 ± 0.9	14.8 ± 0.9	$\mathbf{40.0 \pm 1.0}$	36.0 ± 1.1	16.1 ± 0.8
DiffusionDB-10k	37.7 ± 0.9	37.8 ± 0.8	15.2 ± 0.9	$\mathbf{40.3 \pm 1.2}$	35.4 ± 1.0	15.5 ± 1.1

Table 3: Summary of the most important ablations on ImageNet-100-LT. Average columns are computed over all seven downstream tasks.

Group	Setting	Avg. Lin.	Avg. kNN	Flowers	Pets	IN100-LT
SSL objective	SimCLR	$\mathbf{34.7 \pm 1.2}$	$\mathbf{28.1 \pm 0.6}$	$\mathbf{27.2 \pm 1.5}$	$\mathbf{33.2 \pm 3.3}$	$\mathbf{41.7 \pm 1.3}$
	MoCo	17.8 ± 1.9	20.5 ± 0.8	10.4 ± 3.2	15.0 ± 1.8	26.6 ± 1.9
	DINO	15.2 ± 1.3	10.4 ± 0.4	12.6 ± 4.4	8.7 ± 1.6	20.6 ± 1.1
Generation	Stable Diffusion 3	$\mathbf{41.4 \pm 0.8}$	$\mathbf{32.7 \pm 0.4}$	$\mathbf{37.4 \pm 2.0}$	$\mathbf{41.2 \pm 0.7}$	$\mathbf{51.8 \pm 1.1}$
	FLUX	35.0 ± 1.9	28.8 ± 1.0	29.4 ± 4.6	31.5 ± 2.9	43.7 ± 1.4
	Re-population	32.6 ± 1.7	26.5 ± 0.5	25.7 ± 1.4	31.7 ± 3.8	40.6 ± 0.5
Selection	Mode window	34.6 ± 1.3	28.1 ± 0.6	27.9 ± 3.1	31.6 ± 3.3	41.7 ± 1.3
	Top tail	34.3 ± 1.2	27.3 ± 0.6	27.7 ± 1.0	30.5 ± 1.2	$\mathbf{42.2 \pm 1.4}$
	Uniform	$\mathbf{35.5 \pm 1.2}$	$\mathbf{28.4 \pm 0.6}$	$\mathbf{32.9 \pm 0.4}$	$\mathbf{31.7 \pm 3.9}$	42.0 ± 0.9

First, SimCLR is the strongest SSL objective for the cyclic repair loop. It substantially outperforms MoCo and DINO on both average linear and average kNN transfer. Second, generation quality matters. Stable Diffusion 3 is clearly stronger than FLUX and stronger than the no generation re population baseline, which indicates that the repaired data must be both targeted and semantically plausible. Third, the new MODEWINDOW selector improves over the older TOPTAIL rule. Uniform sampling remains slightly stronger in this isolated selection ablation, which suggests that once the candidate budget is already small, diversity and locality can both be useful. Appendix Figure 3 summarizes these tradeoffs visually.

6 Limitations

BRIDGE is more expensive than fixed data SSL because every cycle requires feature extraction, OOD scoring, and image generation. The successful experiments reported here use a specific compute regime based on A100 nodes and 24 hour wall time limits, and Appendix C shows that total project compute is substantial. The PASS and DiffusionDB experiments are limited to 10k subsets for tractability. The main paper also centers the final comparison on ResNet-50, even though the appendix includes broader architecture ablations. Finally, the submission does not benchmark newer post project methods that appeared after the original draft.

7 Conclusion

The experiments support a simple conclusion. Long tail SSL is not only an optimization problem. It is also a data geometry problem. By repeatedly identifying sparse representation regions and repairing them with frozen diffusion augmentation, BRIDGE improves average transfer across conventional long tailed sources and across two web scale unlabeled sources. The new mode window selector is important because it replaces an overly literal extreme tail rule with a more stable target region. Within the completed experimental protocol, BRIDGE is the strongest average transfer method across all five source regimes that we study.

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A Additional Supporting Figures

B Implementation and Reproducibility

Core training configuration. The completed main runs use ResNet-50 encoders. Most baseline, BRIDGE, and source regime comparisons use 5 cycles with 100 epochs per cycle. The default BRIDGE configuration uses mean kNN distance with $k = 100$, selects 500 samples per cycle, and generates 5 images per selected sample. The default optimizer is AdamW with weight decay 10^{-4}, and method specific learning rates and temperatures follow the completed job configurations.

OOD scoring and selection. Features are extracted without feature normalization and nearest neighbor distances are computed with FAISS using squared ℓ_2 distance. The mode window selector clips the empirical OOD distribution to the 1st to 99th percentile range, builds a histogram with the code level auto bin heuristic, and selects the examples closest to the modal bin center.

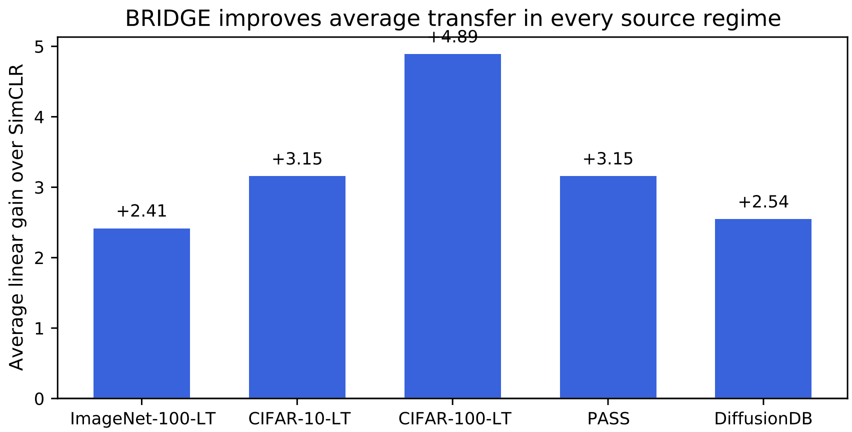

Figure 2: Average linear gain of BRIDGE over source matched SimCLR across all five source regimes. The gain is positive in every case.

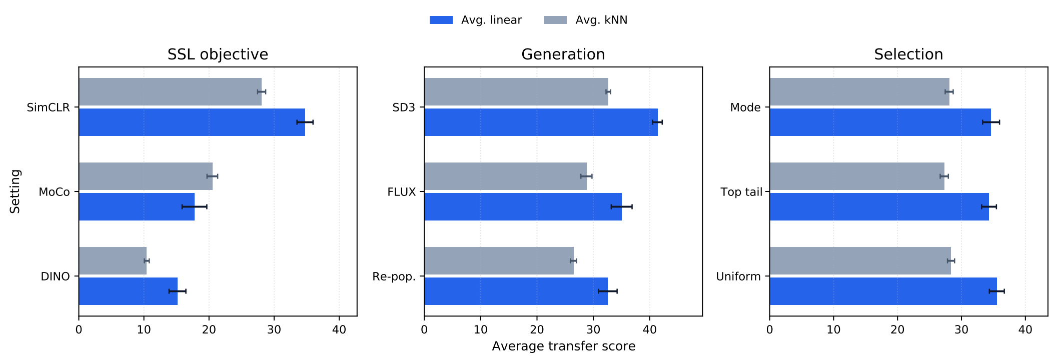

Figure 3: Average linear and average kNN transfer for the main ablation groups. Stable Diffusion 3 is the strongest generator, SimCLR is the strongest SSL objective, and the mode window selector improves over the older top tail rule while uniform sampling remains competitive.

Generators. The main method uses Stable Diffusion 3 in image to image mode. The generation ablation replaces it with FLUX or with a no generation re population baseline. In the completed PASS and DiffusionDB runs we increase the SD3 batch size to 12 to fit the 24 hour Slurm limit.

Source data specifics. PASS and DiffusionDB use `train[:10000]` with pretraining split override `[1.0, 0.0, 0.0]`. This keeps the source data budget aligned across methods while avoiding unnecessary pretraining validation overhead. The long tailed ImageNet-100, CIFAR-10, and CIFAR-100 regimes follow the completed job configurations used in the final sweep.

Seeds and reporting. All main claims are supported by three seed averages. Tables report mean $\pm$ standard deviation over those seeds. The exact table values are generated from the completed job logs by the local export scripts in `scripts/export_paper_results.py` and `scripts/generate_neurips_paper_tables.py`.

C Compute Resources and Runtime

Most completed experiments use one node with two NVIDIA A100 80GB GPUs and a 24 hour wall time limit. The successful DINO ablation uses four A100 80GB GPUs after reducing the number of local crops from six to two. Over the successful experiments reported in the paper, the aggregate training time is approximately 3022 wall clock hours, or about 6142 GPU hours. This excludes

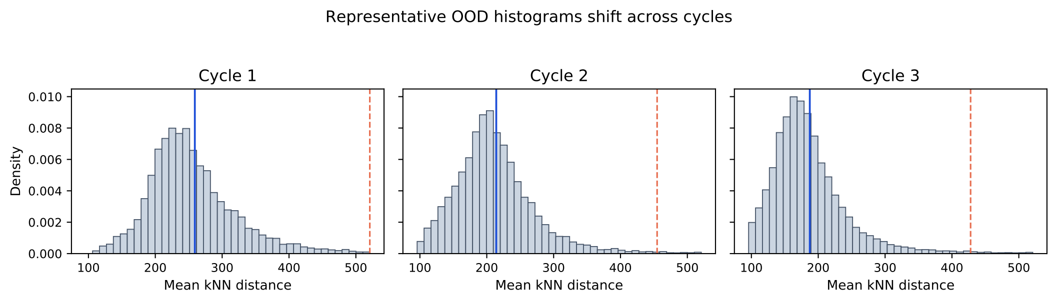

Figure 4: Representative OOD histograms from a successful ImageNet-100-LT TS run. The sparse shoulder moves across cycles, which motivates selecting around a local modal band rather than only the farthest tail.

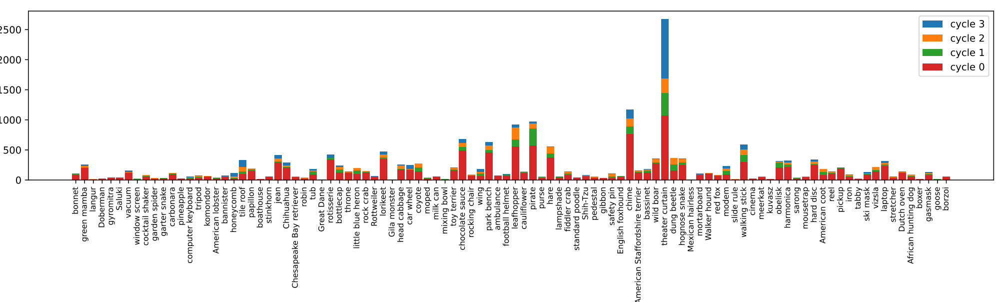

Figure 5: Class count evolution across BRIDGE cycles for a representative ImageNet-100-LT run. This figure is diagnostic only. The pretraining objective itself remains label free.

failed runs, timeout runs, cache migration jobs, and other debugging or pilot jobs, so the full project required more compute than the successful totals alone.

Representative mean wall clock times over three seeds are as follows. On ImageNet-100-LT source pretraining, SimCLR takes 16.9 hours and BRIDGE takes 21.8 hours. On PASS, SimCLR takes 10.2 hours and BRIDGE takes 14.9 hours. On DiffusionDB, SimCLR takes 11.6 hours and BRIDGE takes 16.4 hours. The successful DINO ablation takes 16.3 hours on four GPUs in its final configuration.

D Assets and Terms of Use

We use existing datasets and pretrained generators under their original release terms.

Source datasets. ImageNet-100-LT is derived from ImageNet, whose official access terms restrict the images to non commercial research and educational use. PASS is released under Creative Commons Attribution 4.0. DiffusionDB is released under CC0-1.0. CIFAR-10, CIFAR-100, Stanford Cars, FGVC Aircraft, Oxford Flowers, and Oxford-IIIT Pets are used from their official academic benchmark releases. When an official page provides download terms instead of a standard software style license, we follow those original terms.

Generators. Stable Diffusion 3 Medium is used under the Stability AI Community License. FLUX.1-dev is used only in the ablation and is subject to the FLUX.1-dev Non-Commercial License. We do not redistribute model weights, third party datasets, or generated image corpora as part of this submission.

Table 4: Full linear-probe baseline comparison on ImageNet-100-LT pretraining.

Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Balanced	65.3 ± 1.2	37.0 ± 0.8	8.0 ± 0.5	15.1 ± 0.7	22.4 ± 1.6	$\mathbf{31.2 \pm 2.0}$	39.0 ± 1.0	31.1 ± 1.1
Imbalanced	65.4 ± 4.2	38.6 ± 1.4	7.6 ± 1.1	15.3 ± 0.9	23.7 ± 1.0	28.4 ± 1.3	39.6 ± 0.3	31.2 ± 1.4
BRIDGE (ours)	$\mathbf{70.7 \pm 1.7}$	$\mathbf{43.3 \pm 0.6}$	$\mathbf{9.9 \pm 1.5}$	$\mathbf{16.4 \pm 0.9}$	$\mathbf{26.6 \pm 2.7}$	$\mathbf{31.2 \pm 3.7}$	$\mathbf{42.0 \pm 1.2}$	$\mathbf{34.3 \pm 1.8}$

Table 5: Full kNN baseline comparison on ImageNet-100-LT pretraining.

Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Balanced	63.6 ± 0.7	32.8 ± 0.8	4.4 ± 0.4	9.0 ± 0.4	22.3 ± 0.1	17.2 ± 0.1	34.0 ± 1.1	26.2 ± 0.5
Imbalanced	63.9 ± 0.2	33.2 ± 0.3	4.5 ± 0.6	9.0 ± 0.3	22.5 ± 0.3	16.8 ± 0.2	34.3 ± 0.3	26.3 ± 0.3
BRIDGE (ours)	$\mathbf{66.0 \pm 0.9}$	$\mathbf{35.0 \pm 0.9}$	$\mathbf{5.0 \pm 0.4}$	$\mathbf{9.8 \pm 0.2}$	$\mathbf{24.8 \pm 0.4}$	$\mathbf{18.0 \pm 0.5}$	$\mathbf{35.6 \pm 1.2}$	$\mathbf{27.7 \pm 0.6}$

E Broader Impact

A method that improves representation quality under biased source data can reduce the need for manual relabeling and can make SSL more useful for domains where balanced data is difficult to collect. That is a positive outcome. At the same time, our method relies on large image generators and on web sourced datasets. These carry familiar risks related to license compliance, privacy, and unintended memorization or harmful content. Our experiments stay within existing dataset and model release terms and do not release new generators or scraped image collections. Any future deployment would still require a task specific review of fairness, privacy, and content safety.

F Full Results

G Algorithm

Algorithm 1 BRIDGE training loop

Initialize unlabeled source set $D^{(0)}$, encoder f_θ, frozen generator g_ϕ
for $c = 0, \ldots, C-1$ **do**
  Pretrain f_θ on $D^{(c)}$ for E_c epochs
  Embed all images in $D^{(c)}$ and compute mean kNN distances $\{d_i^{(c)}\}$
  Clip the distance range, build a histogram, and find the modal bin center $m^{(c)}$
  Select B images whose OOD distances are closest to $m^{(c)}$
  Generate G synthetic images per selected image with frozen generator g_ϕ
  Form $D^{(c+1)}$ by adding the generated images to $D^{(c)}$
end for

Table 6: Full linear-probe transfer results across all source datasets and comparison methods.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	73.9 ± 0.5	46.8 ± 0.4	13.7 ± 0.3	19.6 ± 0.2	35.9 ± 7.2	38.6 ± 1.5	50.4 ± 0.5	39.8 ± 1.5
	TS	72.5 ± 0.4	44.0 ± 0.5	12.5 ± 1.2	15.0 ± 0.1	35.7 ± 1.8	40.6 ± 2.2	**56.7 ± 1.2**	39.6 ± 1.1
	SDCLR	41.1 ± 1.7	18.0 ± 0.6	1.4 ± 0.3	2.3 ± 0.5	10.8 ± 1.7	10.4 ± 0.3	21.9 ± 0.6	15.1 ± 0.8
	BRIDGE (ours)	**75.7 ± 0.2**	**48.8 ± 0.3**	**16.1 ± 1.0**	**22.4 ± 0.3**	**36.4 ± 2.4**	**42.7 ± 1.9**	53.7 ± 0.6	**42.3 ± 1.0**
	BRIDGE+TS (ours)	73.2 ± 0.1	45.5 ± 0.4	9.5 ± 1.2	13.0 ± 0.7	26.4 ± 2.7	34.4 ± 1.7	47.6 ± 0.6	35.7 ± 1.0
	BRIDGE+SDCLR (ours)	41.5 ± 0.5	17.7 ± 0.4	1.3 ± 0.3	2.4 ± 0.7	12.1 ± 0.8	10.5 ± 1.9	20.7 ± 1.0	15.2 ± 0.8
CIFAR-10-LT	SimCLR	67.8 ± 0.2	38.0 ± 1.0	6.5 ± 0.4	16.0 ± 0.9	19.2 ± 1.6	24.0 ± 1.3	32.0 ± 0.7	29.1 ± 0.9
	TS	68.5 ± 1.3	36.0 ± 1.7	4.3 ± 0.2	10.3 ± 0.1	13.1 ± 2.0	19.7 ± 2.3	31.5 ± 1.2	26.2 ± 1.3
	SDCLR	39.6 ± 1.1	15.6 ± 0.2	1.1 ± 0.1	2.4 ± 0.9	12.0 ± 1.6	9.1 ± 1.8	16.5 ± 1.6	13.8 ± 1.1
	BRIDGE (ours)	70.7 ± 2.1	41.5 ± 0.6	**8.6 ± 0.4**	**17.0 ± 0.9**	**24.0 ± 3.2**	**29.4 ± 2.1**	**34.4 ± 1.5**	**32.2 ± 1.6**
	BRIDGE+TS (ours)	**73.1 ± 1.0**	**42.0 ± 0.3**	4.4 ± 0.8	12.4 ± 0.5	10.2 ± 1.6	23.2 ± 1.2	34.2 ± 1.1	28.5 ± 0.9
	BRIDGE+SDCLR (ours)	39.1 ± 0.2	15.3 ± 1.0	1.1 ± 0.3	2.1 ± 0.6	8.9 ± 2.6	9.5 ± 1.1	16.8 ± 0.7	13.3 ± 0.9
CIFAR-100-LT	SimCLR	59.0 ± 1.5	31.1 ± 1.6	4.1 ± 0.3	11.2 ± 1.0	12.6 ± 3.4	20.1 ± 1.0	28.7 ± 1.3	23.8 ± 1.4
	TS	56.3 ± 0.2	28.0 ± 0.1	2.8 ± 0.6	8.5 ± 0.2	15.5 ± 2.5	16.2 ± 3.1	26.2 ± 1.8	21.9 ± 1.2
	SDCLR	37.1 ± 0.8	15.7 ± 0.2	1.9 ± 0.1	2.4 ± 0.7	9.4 ± 1.0	9.9 ± 1.7	16.2 ± 0.3	13.2 ± 0.7
	BRIDGE (ours)	**66.8 ± 3.1**	**38.3 ± 2.2**	**6.2 ± 1.8**	**14.6 ± 1.6**	**19.8 ± 0.4**	**22.7 ± 2.3**	**32.6 ± 1.4**	**28.7 ± 1.8**
	BRIDGE+TS (ours)	66.4 ± 3.3	38.0 ± 3.7	4.2 ± 1.0	10.9 ± 0.7	17.2 ± 3.6	21.6 ± 3.3	31.4 ± 2.3	27.1 ± 2.6
	BRIDGE+SDCLR (ours)	39.1 ± 1.9	14.9 ± 0.9	0.9 ± 0.2	3.0 ± 0.4	10.9 ± 2.5	7.9 ± 0.7	16.0 ± 0.3	13.2 ± 1.0
PASS-10k	SimCLR	72.3 ± 0.2	45.5 ± 0.5	11.0 ± 0.2	17.5 ± 1.4	33.3 ± 1.3	33.3 ± 0.9	44.7 ± 0.9	36.8 ± 0.8
	TS	71.5 ± 0.2	43.5 ± 0.4	10.2 ± 1.2	17.2 ± 0.2	33.1 ± 2.1	**36.4 ± 1.8**	49.4 ± 0.4	37.3 ± 0.9
	SDCLR	42.2 ± 1.1	17.7 ± 0.8	1.8 ± 0.3	2.2 ± 0.8	10.6 ± 1.3	8.2 ± 1.6	20.8 ± 0.4	14.8 ± 0.9
	BRIDGE (ours)	**74.2 ± 0.2**	**47.4 ± 0.7**	**13.0 ± 1.2**	**20.4 ± 0.4**	**39.4 ± 3.4**	35.8 ± 0.1	49.5 ± 1.0	**40.0 ± 1.0**
	BRIDGE+TS (ours)	69.1 ± 0.4	40.5 ± 0.8	12.1 ± 0.5	14.3 ± 0.3	32.7 ± 4.1	30.8 ± 1.0	**52.6 ± 0.5**	36.0 ± 1.1
	BRIDGE+SDCLR (ours)	45.5 ± 1.0	19.3 ± 0.2	1.5 ± 0.7	3.3 ± 0.2	11.0 ± 1.6	11.3 ± 1.2	20.9 ± 0.9	16.1 ± 0.8
DiffusionDB-10k	SimCLR	74.4 ± 0.5	48.6 ± 0.2	10.9 ± 0.7	17.6 ± 0.8	31.2 ± 2.4	36.1 ± 1.3	45.3 ± 0.5	37.7 ± 0.9
	TS	72.3 ± 0.3	44.4 ± 0.3	10.7 ± 0.6	15.4 ± 0.6	33.1 ± 1.1	**39.9 ± 2.7**	48.8 ± 0.2	37.8 ± 0.8
	SDCLR	42.4 ± 0.2	18.4 ± 0.2	1.1 ± 0.2	2.4 ± 0.1	13.0 ± 3.7	8.2 ± 1.4	21.0 ± 0.6	15.2 ± 0.9
	BRIDGE (ours)	**76.2 ± 0.1**	**49.8 ± 0.1**	**12.4 ± 1.0**	**20.4 ± 0.2**	**36.1 ± 2.7**	38.0 ± 3.2	48.9 ± 0.8	**40.3 ± 1.2**
	BRIDGE+TS (ours)	68.8 ± 0.1	40.2 ± 0.4	9.2 ± 1.1	12.8 ± 0.5	31.2 ± 2.4	34.8 ± 1.6	**51.0 ± 0.6**	35.4 ± 1.0
	BRIDGE+SDCLR (ours)	42.2 ± 0.7	17.6 ± 0.3	1.1 ± 0.6	2.4 ± 0.1	11.9 ± 3.6	11.7 ± 1.3	21.3 ± 0.8	15.5 ± 1.1

Table 7: Full kNN transfer results across all source datasets and comparison methods.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	68.6 ± 0.4	37.3 ± 0.2	6.0 ± 0.7	11.2 ± 0.4	32.2 ± 0.7	23.6 ± 0.3	43.9 ± 0.9	31.8 ± 0.5
	TS	68.4 ± 0.3	36.5 ± 0.1	4.7 ± 0.2	8.3 ± 0.5	26.6 ± 1.0	23.6 ± 1.7	**52.2 ± 0.5**	31.5 ± 0.6
	SDCLR	32.5 ± 0.4	10.9 ± 0.2	1.0 ± 0.3	2.1 ± 0.1	10.7 ± 0.7	4.7 ± 0.2	11.2 ± 0.2	10.4 ± 0.3
	BRIDGE (ours)	**68.9 ± 0.5**	**37.8 ± 0.1**	**6.9 ± 0.1**	**12.1 ± 0.2**	**36.0 ± 0.1**	**25.3 ± 0.9**	48.2 ± 0.4	**33.6 ± 0.3**
	BRIDGE+TS (ours)	65.2 ± 0.2	33.7 ± 0.5	2.9 ± 0.4	6.6 ± 0.2	18.6 ± 0.4	15.8 ± 1.0	40.5 ± 0.8	26.2 ± 0.5
	BRIDGE+SDCLR (ours)	33.8 ± 0.4	11.5 ± 0.5	1.3 ± 0.4	2.0 ± 0.1	10.2 ± 0.7	4.9 ± 0.2	12.6 ± 0.6	10.9 ± 0.4
CIFAR-10-LT	SimCLR	66.9 ± 0.4	33.4 ± 0.3	3.7 ± 0.4	9.8 ± 0.7	16.5 ± 0.8	13.9 ± 0.8	24.0 ± 0.8	24.0 ± 0.6
	TS	66.3 ± 0.6	27.8 ± 0.4	2.4 ± 0.6	6.4 ± 0.2	10.5 ± 0.7	11.1 ± 0.4	22.0 ± 0.9	20.9 ± 0.5
	SDCLR	30.1 ± 0.3	10.0 ± 0.3	1.7 ± 0.3	2.2 ± 0.2	10.1 ± 0.7	4.4 ± 0.4	9.8 ± 0.3	9.7 ± 0.4
	BRIDGE (ours)	**69.0 ± 1.2**	**35.2 ± 0.9**	**4.2 ± 0.6**	**10.4 ± 0.5**	**17.0 ± 0.6**	**15.2 ± 0.8**	**25.2 ± 0.5**	**25.2 ± 0.7**
	BRIDGE+TS (ours)	67.7 ± 0.8	28.0 ± 0.8	2.1 ± 0.2	5.1 ± 0.7	8.1 ± 1.0	9.1 ± 0.8	21.8 ± 0.5	20.3 ± 0.7
	BRIDGE+SDCLR (ours)	31.3 ± 1.2	10.4 ± 0.5	1.0 ± 0.1	1.8 ± 0.3	9.6 ± 1.3	4.5 ± 0.1	10.1 ± 0.4	9.8 ± 0.6
CIFAR-100-LT	SimCLR	59.3 ± 0.8	30.1 ± 0.9	3.0 ± 0.3	7.4 ± 0.4	13.7 ± 0.2	10.8 ± 0.4	21.8 ± 1.1	20.9 ± 0.6
	TS	53.1 ± 0.6	24.6 ± 0.4	2.4 ± 0.4	5.5 ± 0.3	9.3 ± 0.4	8.3 ± 0.2	18.7 ± 1.0	17.4 ± 0.5
	SDCLR	25.7 ± 1.2	8.5 ± 0.6	0.9 ± 0.4	1.9 ± 0.2	7.3 ± 0.3	3.9 ± 0.3	8.3 ± 1.1	8.1 ± 0.6
	BRIDGE (ours)	**62.5 ± 0.8**	**32.5 ± 0.3**	**3.7 ± 0.3**	**8.7 ± 0.5**	**14.1 ± 0.9**	**12.1 ± 0.7**	**23.0 ± 1.2**	**22.4 ± 0.7**
	BRIDGE+TS (ours)	56.3 ± 1.2	27.8 ± 0.5	1.9 ± 0.0	4.8 ± 0.6	8.0 ± 1.0	9.0 ± 0.6	20.8 ± 0.8	18.4 ± 0.7
	BRIDGE+SDCLR (ours)	29.7 ± 2.2	9.8 ± 1.5	1.4 ± 0.4	2.3 ± 0.2	9.2 ± 1.1	4.8 ± 0.1	8.9 ± 0.6	9.4 ± 0.9
PASS-10k	SimCLR	67.2 ± 0.2	37.0 ± 0.3	**5.4 ± 0.3**	10.3 ± 0.5	30.4 ± 0.0	16.7 ± 0.5	38.3 ± 0.3	29.3 ± 0.3
	TS	65.2 ± 0.7	34.6 ± 0.3	4.4 ± 0.2	7.9 ± 0.2	24.5 ± 0.1	14.4 ± 0.4	39.3 ± 0.4	27.2 ± 0.3
	SDCLR	32.4 ± 1.5	11.5 ± 0.8	1.3 ± 0.2	2.0 ± 0.1	8.6 ± 0.7	5.2 ± 0.2	11.1 ± 0.8	10.3 ± 0.6
	BRIDGE (ours)	**67.8 ± 0.6**	**37.1 ± 0.5**	5.2 ± 0.4	**10.8 ± 0.3**	**33.7 ± 0.6**	**17.9 ± 0.3**	**41.9 ± 0.4**	**30.6 ± 0.5**
	BRIDGE+TS (ours)	59.0 ± 0.4	27.8 ± 0.5	3.3 ± 0.4	6.6 ± 0.0	23.3 ± 0.4	13.9 ± 0.3	39.2 ± 0.3	24.7 ± 0.3
	BRIDGE+SDCLR (ours)	37.9 ± 0.7	13.6 ± 0.0	1.2 ± 0.2	2.7 ± 0.1	8.9 ± 1.3	6.3 ± 0.1	13.9 ± 1.1	12.1 ± 0.5
DiffusionDB-10k	SimCLR	67.3 ± 0.1	36.2 ± 0.4	4.3 ± 0.2	8.8 ± 0.1	26.8 ± 0.4	17.3 ± 0.1	36.7 ± 0.8	28.2 ± 0.3
	TS	64.7 ± 0.4	33.6 ± 0.3	3.1 ± 0.5	7.3 ± 0.3	23.2 ± 0.4	17.0 ± 0.0	37.8 ± 0.6	26.7 ± 0.4
	SDCLR	32.6 ± 1.2	10.0 ± 0.3	1.3 ± 0.1	2.2 ± 0.2	6.1 ± 0.7	4.8 ± 0.4	10.6 ± 0.5	9.7 ± 0.5
	BRIDGE (ours)	**68.1 ± 0.3**	**37.0 ± 0.2**	**5.4 ± 0.5**	**9.5 ± 0.1**	**30.4 ± 0.4**	**18.5 ± 0.1**	**39.9 ± 0.6**	**29.8 ± 0.3**
	BRIDGE+TS (ours)	57.2 ± 0.7	25.6 ± 1.0	2.8 ± 0.3	6.8 ± 0.1	22.3 ± 0.5	15.4 ± 0.6	37.3 ± 0.6	23.9 ± 0.6
	BRIDGE+SDCLR (ours)	37.1 ± 1.3	11.8 ± 0.9	1.3 ± 0.1	2.2 ± 0.2	6.8 ± 0.5	5.9 ± 0.7	12.5 ± 0.7	11.1 ± 0.6

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Table 8: Pretraining objective ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
SimCLR	**71.8 ± 1.2**	**43.5 ± 0.5**	**9.0 ± 0.5**	**16.9 ± 0.3**	**27.2 ± 1.5**	**33.2 ± 3.3**	**41.7 ± 1.3**	**34.7 ± 1.2**
MoCo	46.6 ± 3.1	18.4 ± 2.0	3.1 ± 0.5	4.3 ± 0.8	10.4 ± 3.2	15.0 ± 1.8	26.6 ± 1.9	17.8 ± 1.9
DINO	43.6 ± 0.7	17.8 ± 0.6	1.3 ± 0.5	1.6 ± 0.2	12.6 ± 4.4	8.7 ± 1.6	20.6 ± 1.1	15.2 ± 1.3

Table 9: Pretraining objective ablation with full kNN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
SimCLR	**66.3 ± 0.8**	**35.3 ± 0.4**	**5.1 ± 0.7**	**9.5 ± 0.1**	**24.8 ± 0.6**	**18.9 ± 1.0**	**36.7 ± 0.7**	**28.1 ± 0.6**
MoCo	56.4 ± 1.6	25.9 ± 0.8	3.2 ± 0.4	5.2 ± 0.4	16.3 ± 1.1	12.2 ± 0.7	24.3 ± 0.7	20.5 ± 0.8
DINO	34.4 ± 0.2	12.4 ± 0.1	1.6 ± 0.1	2.4 ± 0.1	6.2 ± 0.8	5.4 ± 0.4	10.7 ± 0.8	10.4 ± 0.4

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- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.

Table 10: Augmentation mechanism ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Stable Diffusion 3	**75.0 ± 0.2**	**48.4 ± 0.2**	**15.1 ± 1.2**	**20.6 ± 0.7**	**37.4 ± 2.0**	**41.2 ± 0.7**	**51.8 ± 1.1**	**41.4 ± 0.8**
FLUX	70.7 ± 0.4	42.1 ± 1.4	10.3 ± 1.3	17.5 ± 0.9	29.4 ± 4.6	31.5 ± 2.9	43.7 ± 1.4	35.0 ± 1.9
Re-population	67.0 ± 2.6	39.4 ± 2.2	8.0 ± 0.6	15.4 ± 0.5	25.7 ± 1.4	31.7 ± 3.8	40.6 ± 0.5	32.6 ± 1.7

Table 11: Augmentation mechanism ablation with full kNN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Stable Diffusion 3	**68.3 ± 0.4**	**36.8 ± 0.2**	**6.5 ± 0.4**	**11.7 ± 0.6**	**35.1 ± 0.4**	**24.2 ± 0.3**	**45.9 ± 0.5**	**32.7 ± 0.4**
FLUX	65.8 ± 1.0	35.4 ± 0.8	5.7 ± 0.7	9.4 ± 0.9	27.7 ± 1.0	18.8 ± 1.3	38.7 ± 1.2	28.8 ± 1.0
Re-population	64.5 ± 0.1	33.4 ± 0.6	4.2 ± 0.6	9.0 ± 0.4	23.0 ± 0.5	17.0 ± 0.4	34.2 ± 1.2	26.5 ± 0.5

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Table 12: Sample-selection strategy ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Mode window	**71.9 ± 0.3**	43.4 ± 0.7	9.0 ± 0.5	16.9 ± 0.2	27.9 ± 3.1	31.6 ± 3.3	41.7 ± 1.3	34.6 ± 1.3
Top tail	70.8 ± 0.5	43.1 ± 1.0	**10.0 ± 1.7**	15.9 ± 1.3	27.7 ± 1.0	30.5 ± 1.2	**42.2 ± 1.4**	34.3 ± 1.2
Uniform	71.5 ± 0.2	**43.8 ± 0.7**	9.3 ± 0.9	**17.5 ± 1.2**	**32.9 ± 0.4**	**31.7** ± 3.9	42.0 ± 0.9	**35.5 ± 1.2**

Table 13: Sample-selection strategy ablation with full kNN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Mode window	**66.3 ± 0.8**	35.3 ± 0.4	**5.1 ± 0.7**	9.5 ± 0.1	24.8 ± 0.6	18.9 ± 1.0	36.7 ± 0.7	28.1 ± 0.6
Top tail	65.7 ± 0.8	34.4 ± 0.2	4.1 ± 0.4	9.6 ± 0.2	23.9 ± 0.9	18.1 ± 0.8	35.4 ± 1.2	27.3 ± 0.6
Uniform	66.1 ± 1.0	**35.7 ± 0.3**	4.9 ± 0.1	**10.2 ± 0.7**	**25.3 ± 0.4**	**19.0 ± 0.8**	**37.4 ± 0.6**	**28.4 ± 0.6**

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Table 14: Cycle count ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
2 cycles	67.9 $\pm$ 0.6	40.7 $\pm$ 0.6	9.4 $\pm$ 1.8	16.0 $\pm$ 1.5	**30.5** $\pm$ **3.2**	27.4 $\pm$ 1.7	40.1 $\pm$ 0.4	33.1 $\pm$ 1.4
10 cycles	70.5 $\pm$ 1.2	43.8 $\pm$ 0.6	9.2 $\pm$ 0.9	17.2 $\pm$ 1.6	24.8 $\pm$ 5.2	31.3 $\pm$ 1.7	41.9 $\pm$ 0.6	34.1 $\pm$ 1.7
20 cycles	**71.6** $\pm$ **1.1**	**44.6** $\pm$ **0.9**	9.8 $\pm$ 0.3	**17.3** $\pm$ **1.2**	27.0 $\pm$ 2.1	**33.3** $\pm$ **4.4**	**42.1** $\pm$ **1.5**	**35.1** $\pm$ **1.6**
5 cycles	70.7 $\pm$ 1.7	43.3 $\pm$ 0.6	**9.9** $\pm$ **1.5**	16.4 $\pm$ 0.9	26.6 $\pm$ 2.7	31.2 $\pm$ 3.7	42.0 $\pm$ 1.2	34.3 $\pm$ 1.8

Table 15: Cycle count ablation with full kNN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
2 cycles	65.1 $\pm$ 0.1	34.1 $\pm$ 0.5	4.6 $\pm$ 0.3	9.3 $\pm$ 0.3	23.8 $\pm$ 0.3	17.3 $\pm$ 0.7	34.5 $\pm$ 0.6	26.9 $\pm$ 0.4
10 cycles	66.4 $\pm$ 0.5	35.4 $\pm$ 0.3	4.8 $\pm$ 0.3	**10.4** $\pm$ **0.6**	**25.2** $\pm$ **0.8**	**18.6** $\pm$ **0.9**	36.0 $\pm$ 0.7	**28.1** $\pm$ **0.6**
20 cycles	**66.5** $\pm$ **0.7**	**35.6** $\pm$ **0.8**	4.8 $\pm$ 0.3	9.5 $\pm$ 0.4	24.9 $\pm$ 0.5	18.6 $\pm$ 1.2	**36.3** $\pm$ **1.2**	28.0 $\pm$ 0.7
5 cycles	66.0 $\pm$ 0.9	35.0 $\pm$ 0.9	**5.0** $\pm$ **0.4**	9.8 $\pm$ 0.2	24.8 $\pm$ 0.4	18.0 $\pm$ 0.5	35.6 $\pm$ 1.2	27.7 $\pm$ 0.6

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Table 16: Backbone architecture ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ResNet-18	67.7 ± 0.8	40.4 ± 0.5	**10.6 ± 0.7**	15.4 ± 0.3	26.1 ± 2.4	29.5 ± 2.7	41.0 ± 1.2	32.9 ± 1.2
ResNet-50	**71.7 ± 1.0**	**44.0 ± 1.1**	9.0 ± 0.5	**17.4 ± 0.7**	28.8 ± 1.7	**32.0 ± 1.6**	41.7 ± 1.3	**34.9 ± 1.1**
ViT-S	62.7 ± 0.4	37.6 ± 0.6	8.8 ± 0.5	12.2 ± 0.2	35.1 ± 4.2	30.2 ± 1.6	42.5 ± 1.2	32.7 ± 1.3
ViT-B	66.0 ± 2.3	40.6 ± 2.3	8.3 ± 0.5	14.3 ± 0.8	**38.8 ± 2.6**	31.3 ± 2.1	**44.5 ± 1.7**	34.8 ± 1.8

Table 17: Backbone architecture ablation with full kNN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ResNet-18	65.0 ± 0.3	34.4 ± 0.3	4.7 ± 0.3	**9.6 ± 0.4**	23.5 ± 1.4	16.6 ± 0.3	36.3 ± 1.1	27.2 ± 0.6
ResNet-50	66.3 ± 0.8	35.3 ± 0.4	5.1 ± 0.7	9.5 ± 0.1	24.8 ± 0.6	18.9 ± 1.0	36.7 ± 0.7	28.1 ± 0.6
ViT-S	67.3 ± 0.4	37.5 ± 0.3	5.3 ± 0.2	7.6 ± 0.1	36.2 ± 0.7	20.1 ± 1.0	42.1 ± 1.0	30.9 ± 0.5
ViT-B	**69.3 ± 0.8**	**40.1 ± 0.8**	**6.0 ± 0.2**	8.3 ± 0.2	**39.8 ± 0.8**	**21.9 ± 1.4**	**45.1 ± 0.7**	**32.9 ± 0.7**

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